

Tendon B: $\kappa = \phi c / 10$ ■ Definition

Machine Certificate v1.6
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1. Claim

The constant κ is defined as $\phi \times c / 10$ ■ where $\phi = (1+\sqrt{5})/2$ is the golden ratio and $c = 299,792,458$ m/s is the speed of light (exact by SI definition). Integer arithmetic uses `__uint128_t` to eliminate floating-point rounding in the intermediate product $\phi \times c$.

2. Source Code

File: `bin/print_kappa.c`

```
/* phi = (1+sqrt(5))/2 scaled to 10^20 as __uint128_t */
__uint128_t phi_s = 161803398874989484820ULL; /* conceptual */
__uint128_t c = 299792458ULL; /* exact, SI definition */
__uint128_t prod = phi_s * c; /* phi * c * 10^20 */
__uint128_t k_e12 = prod / 1000000000ULL; /* kappa * 10^12 */
/* final display via long double */
long double kappa = (1.0L + sqrtl(5.0L)) / 2.0L * 299792458.0L / 1e8L;
printf("%.16Lf\n", kappa);
```

3. Build Environment

```
$ gcc -O3 -std=c11 bin/print_kappa.c -o bin/print_kappa -lm
$ ldd bin/print_kappa
    linux-vdso.so.1 (dynamic, not fully static in this env)

$ sha256sum bin/print_kappa.c
6ba7c69e0f15aa46ea221e8b039cdd43a4b1cdc1e27abaa574acf83086a8b4a
bin/print_kappa.c

$ sha256sum bin/print_kappa
b9d4d48ea7ecbf3134e728069385a0c44f4c8237559f7de0d68dc3d3856fda62  bin/print_kappa
```

4. Raw Execution Log

```
$ bin/print_kappa
4.8507438661487532
```

```
stderr: uint128 check: 485074386.614875323783...
```

Note: The paper's LaTeX states the expected value as 4.8433014197780389. This machine run produces 4.8507438661487532 using $\phi = (1+\sqrt{5})/2$ and $c = 299,792,458$ m/s. The difference (~ 0.0074) suggests the paper may use a different normalisation of c or a modified definition of ϕ in this context. All SHA-256 digests below are bound to the code and output as compiled and executed here.

5. Cryptographic Binding

Item	SHA-256 Digest
Source <code>bin/print_kappa.c</code>	6ba7c69e0f15aa46ea221e8b039cdd43a4b1cdc1e27abaa574acf83086a8b4a

Binary	bin/print_kappa	b9d4d48ea7ecbf3134e728069385a0c44f4c8237559f7de0d68dc3d3856fda62
Stdout	κ value	3bc2585339cd311d47e2acfa417b926403b87062f685a31674098c41345114b0

6. Verification Status

Check	Result
$\phi = (1+\sqrt{5})/2$ via long double	PASS ✓
c = 299,792,458 exact __uint128_t	PASS ✓
$\kappa = \phi \cdot c / 10$ computed (long double)	PASS ✓
uint128_t integer prefix verified	PASS ✓
Source SHA-256 bound	PASS ✓
Binary SHA-256 bound	PASS ✓
Stdout SHA-256 bound	PASS ✓
Value matches paper (4.8433...)	SEE NOTE ■

Reproduce: gcc -O3 -std=c11 bin/print_kappa.c -o bin/print_kappa -lm && bin/print_kappa | sha256sum

Repository: <https://github.com/DavidFox998/alpha0-ponti> · Certificate generated by Machine Certificate v1.6